MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES 6SJ7, 6SJ7WGT, AND 12SJ7
a b c

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, RF sharp cutoff

Outline:	a,c	---	8-1 (EIA)
	b	---	9-12 (EIA)
Base:	a,c	---	B8-21 (phenolic)
	b	---	B8-26 (phenolic)
Envelope:	a,c	---	MT8
	b	---	T9
Cathode:	a,b, c	---	Coated unipotential

Base connections:

Pin No.	---	All		1	2	3	4	5	6	7	8
Element	---	a,c	---	sh	h	g3	g1	k	g2	h	a
		b	---	Base sleeve	h	g3	g1	k	g2	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:			Ef V	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ec3 Vdc	Pg2 W	Pp W	Ehk v	Alt ft (Note 2)
	a,b	(Triode)	6.3 ±10%	275	---	---	---	---	2.8	100	---
	c	(Triode) (see note 1)	12.6 ±10%	275	---	---	---	---	2.8	100	---
	a,b	(Pentode)	6.3 ±10%	330	---	140	---	0.7	2.8	100	---
	c	(Pentode)	12.6 ±10%	330	---	140	---	0.7	2.8	100	---
TEST CONDITIONS:	a,b		6.3	250	-3	100	0	---	---	---	---
	c		12.6	250	-3	100	0	---	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1031	Vibration fatigue	b		---	---	---	---	---	---
---	Vibration fatigue - test end points:								
1336	Heater-cathode leakage	b		---	---	Ihk	---	30	µAdc
1256	Electrode current (grid)	b		---	---	Ic1	---	-1.0	µAdc
1306	Transconductance	b		---	---	Sm	1,070	---	µmhos
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)	All		0.65	II	Ib	2.0	4.0	mAdc
1266	Total grid current	All	See note 3	0.65	II	Ic1	0	-0.5	µAdc
1306	Transconductance	All		0.65	II	Sm	1,325	1,975	µmhos
1231	Emission	All	Eb = Ec1 = Ec2 = Ec3 = 30 Vdc (see note 3)	0.65	II	Is	60	---	mAdc
1201	Short and discontinuity detection	All		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	All	Rp = 10,000 ohms	---	---	Ep	---	100	mVac
1301	Heater current	a,b		---	---	If	275	325	mA
1301	Heater current	c		---	---	If	138	162	mA
1336	Heater-cathode leakage	All		---	---	Ihk	---	20	µAdc
1256	Electrode current (2) (anode)	All	Ec1 = -10 Vdc	---	---	Ib	0	30	µAdc
1256	Electrode current (screen)	All		---	---	Ic2	0.4	1.2	mAdc
1331	Direct-interelectrode capacitance	All	Shield No. 308 (T9 bulbs)	---	---	Cga	---	0.005	pF
		a,c	Without shield			Cin	4.9	7.1	pF
		b	Shield No. 308			Cin	4.9	7.8	pF
		a,c	Without shield			Cout	5.2	8.8	pF
		b	Shield No. 308			Cout	5.2	9.4	pF
1101	Secureness of base, cap, or insert	All		---	---	---	---	---	---
1111	Base pin solder depth (rigid leads)	b		---	---	---	---	---	---
1211	Insulation of electrodes	All		4.0	S3	---	---	---	---
1131	Metallic base sleeve quality	b		---	---	---	---	---	---
1041	Shock	b	450 G	15	S2	---	---	---	---
---	Shock test end points:								
1031	Low-frequency vibration	b		---	---	Ep	---	150	mVac
1336	Heater-cathode leakage	b		---	---	Ihk	---	30	µAdc
1266	Total grid current	b		---	---	Ic1	---	-1.0	µAdc
1306	Transconductance	b		---	---	Sm	1,070	---	µmhos
1105	Permanence of marking	All		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3								
1501	Intermittent life	All	Group A; Ehk = 100 V; Eb = Ec2 = Ec3 = 250 Vdc; Ec1 = -8 Vdc; Rg1 = 0.1 Meg (min)	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours):								
1306	Transconductance	All		---	---	Sm	1,070	---	µmhos

NOTES:

1. Tie g2 and g3 to anode.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3094)

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*.

4. REMARKS

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